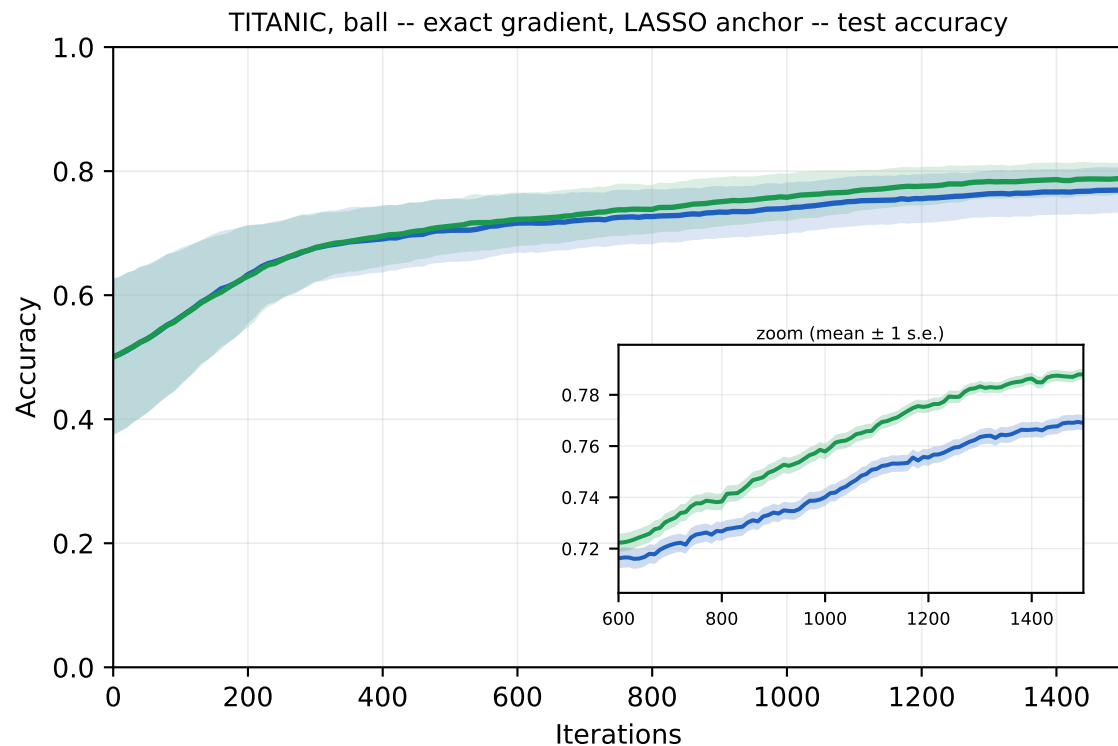
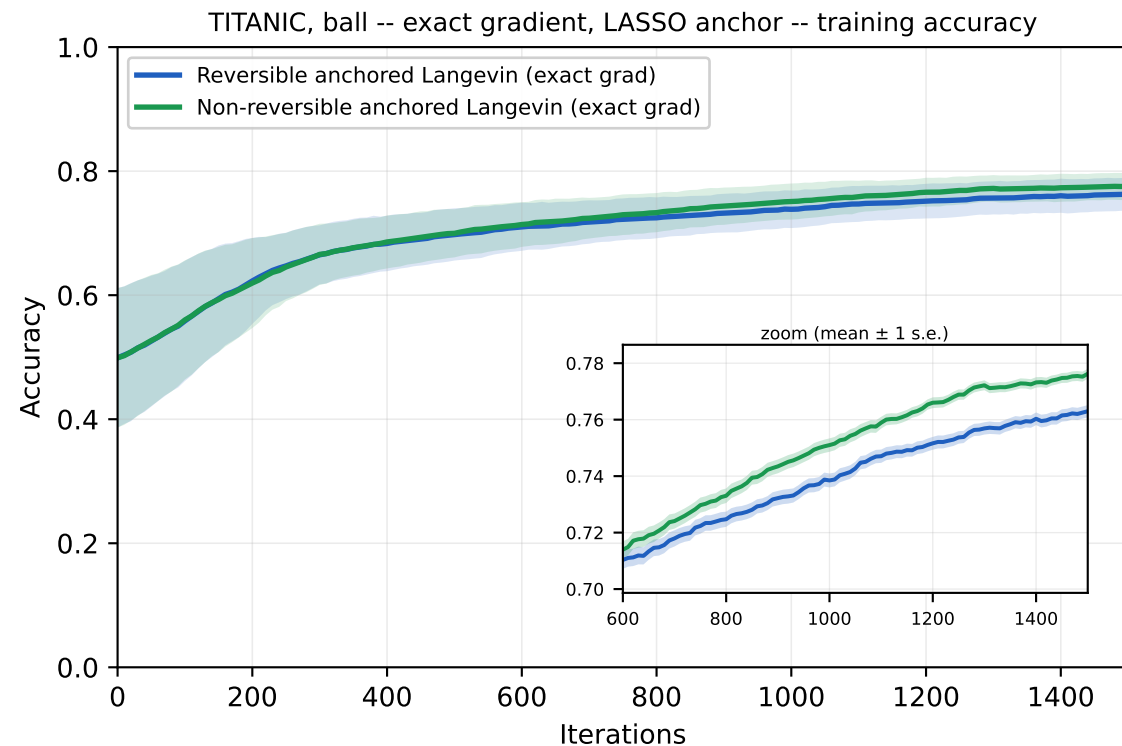


titanic_ball: exact-gradient anchored Langevin, s = (5, 5, 5)



TITANIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient (no mini-batching). $\sigma = 10$, $\lambda_{\text{lasso}} = 7.12 = 0.01 n_{\text{train}}$, $\delta = 0.0108$ (so a in $[0.50, 1]$). Block strengths $s = (5, 5, 5)$, acting on $w_{1..9}$ only. Step $\eta = 1e-05$ (actually used). $R = 150$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval. η and s selected by a TRAINING-ONLY sweep over 5 step sizes $\times$ 4 strengths; these curves are a CONFIRMATION run on independent replicates (seed 4100) not used for that selection. Verdict on the paired training difference: non-reversible WINS.